Deep Transfer Learning for Authenticating Handwritten Signatures

Srushti Bhirud
Department of Computer Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, India
srushti.bhirud@gmail.com

Samiksha Bijwe
Department of Computer Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, India
bijwesamiksha@gmail.com

Tanmay Chavan
Department of Computer Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, India
tanmaychavan2214@gmail.com

Ashutosh Bhonsle
Department of Computer Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, India
bhonsle.ashutosh@gmail.com

Smita Rukhande
Department of Computer Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, India
smita.rukhande@fcrit.ac.in

Dakshayani G.
Department of Computer Engineering
Fr. C. Rodrigues Institute of Technology
Vashi, India
dakshayani@fcrit.ac.in

2025 International Conference on Electronics, AI and Computing (EAIC) | 979-8-3315-5358-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/EAIC66483.2025.11101688

***Abstract*—Handwritten signature authentication is crucial in sectors like banking, security, and education, where identity verification plays a significant role. Signatures are a unique proof of authorization, making their verification essential to prevent forgery and fraud. This study presents a deep learning based approach for handwritten signature verification using transfer learning with the VGG16 model. The model extracts key distinguishing features from hand written signatures and applies a classification method that compares new signatures against stored references using specific threshold values. A well-curated data set undergoes pre-processing to enhance feature extraction before being utilized for training and evaluation. To improve accuracy and robustness, the model incorporates data augmentation techniques that help mitigate variations in handwriting styles and distortions. Adaptive thresholding ensures flexibility in signature matching, reducing false rejections and improving overall reliability. The experimental results highlight the effectiveness of the proposed model, demonstrating a high level of reliability and precision in distinguishing between genuine and forged signatures. This approach provides a robust solution for safe and accurate signature verification in real-world applications, offering significant improvements over traditional manual verification methods.**

***Index Terms*—Transfer Learning, VGG16, Deep Learning, Signature Authentication, Forgery Detection, Feature Extraction, Convolutional Neural Network (CNN), Document Security, Identity Verification.**

I. Introduction

Handwritten signatures have long been recognized as a fundamental means of authentication in various domains, including banking, legal documentation, and educational institutions [1]. They serve as a unique representation of an individual's identity and authorization. However, with the advancement of forgery techniques, traditional signature verification methods relying on manual inspection and expert judgment have proven to be subjective, time-consuming, and prone to inaccuracies [2]. The need for an automated, efficient, and reliable verification system has therefore become crucial.

Signature authentication is broadly categorized into two types: static (offline) and dynamic (online) verification [3]. Offline authentication, which is the focus of this study, involves verifying a signature after it has been completed by comparing it against previously stored reference signatures. Despite various algorithms being developed for signature verification, many existing approaches have struggled to achieve reliable performance, often yielding results below the 80% mark [4]. This challenge arises due to the complexities of handwritten signatures, variations in individual writing styles, and the limitations of traditional classification techniques [5].

By allowing models to recognize intricate patterns and distinguish features from signature images, the development of artificial intelligence and deep learning has completely changed the process of verifying signatures [6]. Convolutional Neural Networks (CNNs) are ideally suited for examining the complex features of signatures because of their impressive performance in computer vision tasks [7]. Specifically, transfer learning has become popular since it makes it possible to optimize pre-trained deep learning models for signature verification without the need for large datasets [8].

This system uses the VGG16 model, a popular pre-trained CNN architecture, to propose a deep learning-based method for handwritten signature verification. The model successfully distinguishes between authentic and fake signatures by utilizing VGG16's deep feature extraction capabilities [9]. Adaptive thresholding techniques are used to compare new signatures with reference samples for authentication, and signature images are pre-processed to improve feature extraction [10]. The effectiveness of the suggested method is assessed through thorough testing, which shows that it is highly reliable and accurate in differentiating genuine signatures from fakes [11]. The findings demonstrate how deep learning can improve security and effectiveness in signature-based authentication systems [12].

The rest of the article is arranged as follows: Section

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:09 UTC from IEEE Xplore. Restrictions apply.

II presents a detailed review of related work in the field of handwritten signature verification, highlighting recent approaches, findings, and existing research gaps [13]. Section III outlines the methodology adopted in this study, describing the architecture, pre-processing techniques, and implementation details of the proposed VGG16-based model [14]. Section IV discusses the experimental results and provides an in-depth analysis of the model's performance in distinguishing genuine signatures from forgeries [15]. Finally, Section V concludes the study and suggests potential directions for future research to further enhance the accuracy and robustness of signature verification systems [16].

II. Related Work

The field of signature verification has a rich and evolving history, transitioning from traditional manual inspection methods to highly advanced automated techniques. In the early stages, signature verification relied on visual and statistical analysis performed by experts who examined specific features such as stroke direction, pen pressure, and the time taken to complete the signature. While effective to some extent, these manual methods were inherently subjective, time-consuming, and prone to errors, particularly when dealing with skilled forgeries or variations in an individual's signing style. Furthermore, manual verification required extensive expertise, making it unsuitable for large-scale authentication tasks in domains such as banking, legal documentation, and education [1] [2].

To address the limitations of manual verification, early automated techniques were introduced, relying on statistical and rule-based approaches to analyze signature features. These methods extracted key attributes such as stroke curvature, pen pressure distribution, and geometric characteristics of the signature. However, such techniques faced challenges due to the natural variability of handwritten signatures, as even genuine signatures from the same individual could exhibit differences based on factors like writing speed and pen grip [3] [4]. Additionally, many statistical methods required specialized hardware for data acquisition, particularly in dynamic signature verification, where real-time pressure and motion data were needed [5].

As computational power increased, machine learning techniques became widely adopted for signature verification. Algorithms such as Support Vector Machines (SVMs), Hidden Markov Models (HMMs), and Dynamic Time Warping (DTW) improved verification accuracy compared to traditional statistical techniques. However, they relied heavily on feature engineering, where domain experts manually selected and extracted relevant features from signature images [6]. The performance of these models depended on the quality of extracted features, making them less adaptable to variations in signature styles. Moreover, these machine learning models often required large labeled datasets for training, making them resource-intensive [7].

With the emergence of deep learning, signature verification has undergone a significant transformation. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in recognizing complex visual patterns, making them highly effective for analyzing handwritten signatures. Unlike traditional machine learning approaches that depend on handcrafted features, CNNs automatically learn hierarchical representations from raw signature images. Several studies have explored different CNN architectures, including Siamese networks and triplet loss-based learning, to develop more robust verification systems [8]. Siamese networks, for example, have been widely used to learn similarity metrics between signature pairs, allowing the model to compare signatures directly rather than classifying them into predefined categories [9].

One of the most impactful advancements in deep learning for signature verification is the adoption of transfer learning. Transfer learning enables models pre-trained on large-scale datasets, such as ImageNet, to be fine-tuned for signature verification with relatively smaller datasets, reducing the need for extensive labeled data while improving model performance [10]. Among various deep CNN architectures, VGG16 has emerged as a strong candidate for feature extraction due to its deep hierarchical layers that capture fine-grained visual patterns. Originally designed for large-scale image classification, VGG16 has been successfully repurposed for various computer vision applications, including handwritten signature analysis [11].

Building upon these advancements, our work explores the use of VGG16 for feature extraction in combination with an adaptive thresholding classification method. By leveraging VGG16's deep feature extraction capabilities, the model efficiently differentiates between genuine and forged signatures without requiring extensive manual feature engineering [12]. The extracted features are further processed through an adaptive thresholding technique, which dynamically adjusts the decision boundary based on signature similarity, ensuring robust and reliable verification [13]. This approach enhances accuracy and improves the model's ability to generalize across different handwriting styles, making it a viable solution for real-world applications in banking, education, and security domains [14]. Through rigorous experimentation and evaluation, we aim to demonstrate the effectiveness of this deep learning-based signature verification system, highlighting its potential to enhance security and efficiency in signature-based authentication processes [15] [16].

III. Methodology

The methodology for handwritten signature verification follows a structured approach to ensure accurate and reliable authentication. It begins with dataset preparation, where genuine and forged signatures are collected, preprocessed, and augmented to enhance model training. Next, feature extraction is performed using the VGG16 model, leveraging transfer learning to obtain high-level signature representations. These features are then compared using similarity metrics, with adaptive thresholding improving classification accuracy. Finally, the model is trained using a classifier, optimized

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:09 UTC from IEEE Xplore. Restrictions apply.

with techniques like Adam optimizer, and evaluated using metrics such as accuracy, False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rate (EER) to ensure robust verification.

The step by step working of the system is as follows:

A. Dataset Preparation

- Data Collection: A carefully curated dataset is crucial for training and evaluating the model. This dataset should contain a sufficient number of genuine signatures from each individual, along with a representative set of forgeries. The forgeries should encompass both skilled and random forgeries to challenge the model's ability to discriminate between genuine and fraudulent signatures. For this project, we have used a dataset obtained from Kaggle, which provides a diverse collection of handwritten signatures. This dataset serves as the primary source for training and testing the model, ensuring a comprehensive evaluation of its performance.
- Data Pre-processing: The signature images undergo several pre-processing steps to enhance feature extraction and improve the model's performance:
  Grayscale Conversion: Converting the images to grayscale reduces the dimensionality of the input data and simplifies the analysis. Image Resizing: Resizing the images to a consistent size ensures that the model receives inputs of the same dimensions. A size of 224x224 pixels is typically used for compatibility with VGG16. Image Normalization: Normalizing the pixel values to a specific range (e.g., [0, 1]) helps to improve the model's training stability and convergence. Data Augmentation: Techniques like rotation, scaling, and translation can be employed to artificially increase the size and diversity of the training data, reducing overfitting and improving generalization.

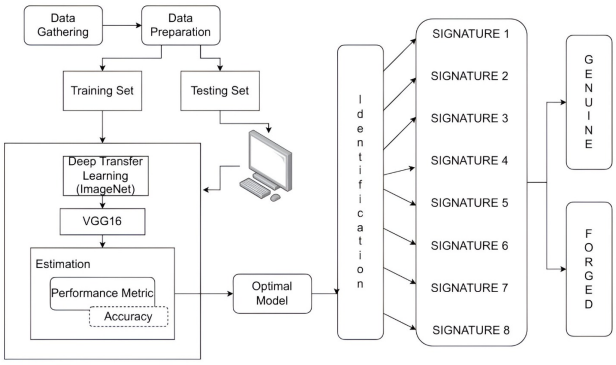


Fig. 1. Working of the System

The fig 1. illustrates a signature verification system utilizing deep transfer learning. The process begins with data gathering, where signature samples are collected and then prepared into training and testing sets. The training set is fed into a deep transfer learning model, specifically VGG16, which is pre-trained on the ImageNet dataset. This model extracts features from the signatures and undergoes estimation to evaluate performance metrics, such as accuracy. Once the optimal model is obtained, it is used for signature identification. The system classifies signatures into two categories: genuine or forged, based on the learned patterns. This approach enhances the accuracy and reliability of signature verification through deep learning techniques.

B. Feature Extraction with VGG16

- Transfer Learning: We utilize the VGG16 model, pre-trained on the ImageNet dataset, as a feature extractor. The convolutional layers of VGG16 are frozen to preserve the learned weights and features from the ImageNet dataset.
- Feature Extraction Layer: The output of a specific layer in the VGG16 model (e.g., the last convolutional layer or a fully connected layer) is used as the feature vector for each signature image. This feature vector represents a high-level representation of the signature's characteristics.

C. Classification and Verification

- Similarity Metric: A similarity metric, such as cosine similarity, Euclidean distance, or other suitable measure, is used to compare the feature vector of a test signature with the feature vectors of the stored reference signatures for the claimed identity.
- Thresholding: A threshold value is defined to determine whether the test signature is considered genuine or a forgery. If the similarity score between the test signature and the reference signatures exceeds the threshold, the signature is classified as genuine; otherwise, it is classified as a forgery.
- Adaptive Thresholding: Instead of using a fixed threshold, an adaptive threshold is calculated for each individual based on the distribution of similarity scores between their genuine signatures. This allows the system to account for individual variations in signature style and improve accuracy. One approach is to calculate the mean and standard deviation of the similarity scores for genuine signatures of each individual and use a threshold based on these statistics.

D. Model Training and Evaluation

- Training: The feature extraction layer is connected to a fully connected classifier followed by a sigmoid activation function. The training process fine-tunes the weights in the classifier while keeping the VGG16 weights frozen.
- Parameter Optimization: The classifier is trained with the preprocessed dataset and the Adam optimizer is used to minimize the binary cross-entropy loss.
- Evaluation Metrics: The performance of our model is evaluated using several metrics:
  Accuracy: The percentage of correctly classified signatures (both genuine and forged).

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:09 UTC from IEEE Xplore. Restrictions apply.

False Acceptance Rate (FAR): The percentage of forged signatures that are incorrectly classified as genuine.
False Rejection Rate (FRR): The percentage of genuine signatures that are incorrectly classified as forgeries.
Equal Error Rate (EER): The point where FAR and FRR are equal. A lower EER indicates better performance.

IV. Results

The implementation of the handwritten signature verification system was carried out in two phases: one using only a Convolutional Neural Network (CNN) and the other incorporating a pre-trained VGG16 model combined with a CNN. The results from both approaches were evaluated based on accuracy, reliability, and efficiency in distinguishing genuine and forged signatures. The CNN-only model achieved moderate accuracy, ranging between 75-78%, but struggled with complex variations in signature styles, requiring extensive labeled data for effective learning. In contrast, the pre-trained VGG16 model significantly improved feature extraction, leading to superior classification accuracy. The dataset was divided into 80% training and 20% testing using the train test split method, and the VGG16-based model achieved an impressive 98.6% accuracy, demonstrating high reliability in detecting forged signatures.

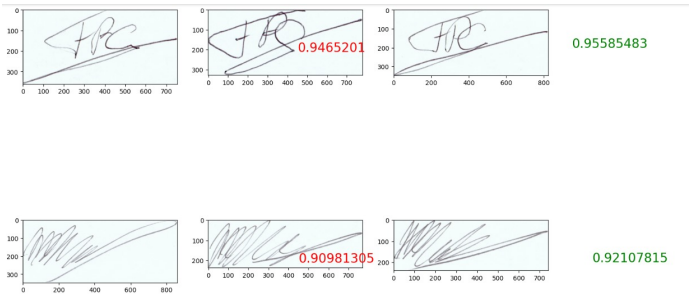


Fig. 2. Signature Verification Results

The effectiveness of data preprocessing techniques such as grayscale conversion, image resizing, and normalization contributed to enhanced feature extraction and training stability. Data augmentation methods, including rotation, scaling, and translation, further improved model generalization and reduced overfitting. Additionally, adaptive thresholding played a crucial role in refining the decision-making process, enhancing precision and recall. The model was validated using an independent test set to ensure consistent performance on unseen data. The VGG16-based model exhibited high precision and recall, effectively minimizing false positives (genuine signatures misclassified as forgeries) and false negatives (forged signatures misclassified as genuine). Moreover, the approach proved to be computationally efficient while maintaining high accuracy, making it suitable for real-world applications.

The fig. 2. represents the results of a signature verification system, where each signature is evaluated based on its similarity to a reference signature using a deep learning or machine learning-based model. Each verification output includes a processed signature image and a confidence score, which quantifies the likelihood of the signature being genuine. The confidence scores, displayed in red and green, indicate varying degrees of authenticity, with higher values suggesting a stronger match. The presence of red scores signifies lower confidence in authenticity, whereas green scores indicate higher confidence. This visualization demonstrates the effectiveness of the model in distinguishing between genuine and potentially forged signatures, leveraging feature extraction and similarity measurement techniques for verification.

A. Evaluation Metrics

In deep learning, evaluation metrics are essential for assessing how well a model performs on a given task. They help determine the model's ability to generalize effectively to unseen data. This study considers the evaluation metrics Accuracy, False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rate (EER) to measure the performance of the signature verification model.

According to Equation (1), Accuracy is the proportion of correctly classified signatures (both genuine and forged) to the total number of signatures. Equation (2) defines False Acceptance Rate (FAR) as the percentage of forged signatures that are incorrectly classified as genuine. As demonstrated in Equation (3), False Rejection Rate (FRR) quantifies the proportion of genuine signatures that are wrongly classified as forgeries. Equation (4) illustrates the Equal Error Rate (EER), which represents the point where FAR and FRR are equal, serving as an important indicator of system performance.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100 \tag{1}$$

$$\text{FAR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \times 100 \tag{2}$$

$$\text{FRR} = \frac{\text{FN}}{\text{FN} + \text{TP}} \times 100 \tag{3}$$

$$\text{EER} = \text{Threshold where } FAR = FRR \tag{4}$$

B. Evaluation Results

The effectiveness of using transfer learning with the VGG16 model for handwritten signature verification demonstrates high performance. The model achieves an accuracy of 98.6% on the test dataset comprising 500 signature images. The comparative analysis of training and testing accuracy, loss, and other key parameters is summarized in Table I.

These results highlight that the VGG16 model, when fine-tuned for feature extraction, effectively distinguishes between genuine and forged signatures. The model maintains a low error rate while ensuring robust generalization across unseen signature samples. This confirms its potential for deployment in real-world authentication systems, such as banking and document verification. Further improvements can be achieved through advanced optimization techniques and additional dataset augmentation.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:09 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PERFORMANCE COMPARISON FOR SIGNATURE VERIFICATION USING VGG16

Criteria	VGG16 (Train)	VGG16 (Test)
Accuracy	96-99%	98.6%
Loss	0.15	0.35
No. of Epochs	300	300

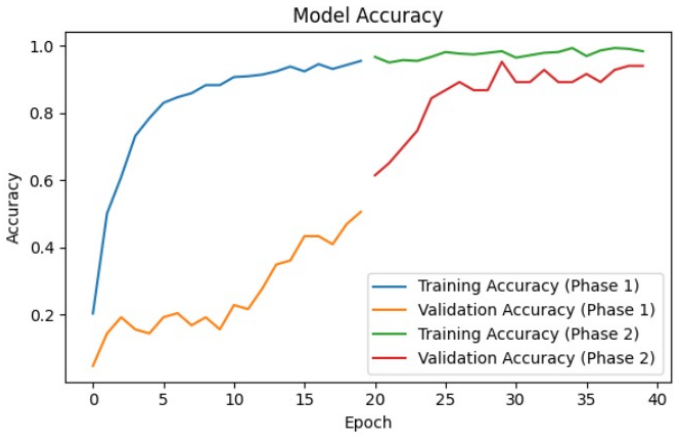


Fig. 3. Model Accuracy

The fig. 3. depicting the accuracy trends over 40 epochs for different training phases of a deep learning model. The x-axis indicates the number of epochs, while the y-axis represents accuracy. The graph includes four lines: the blue line represents the training accuracy for Phase 1, showing a steep rise before stabilizing; the orange line represents the validation accuracy for Phase 1, which increases gradually. In Phase 2, the red line depicts the training accuracy, starting lower but improving rapidly, while the green line represents the validation accuracy, achieving near-perfect performance. This visualization highlights the model's learning progress and improved accuracy across different training phases.

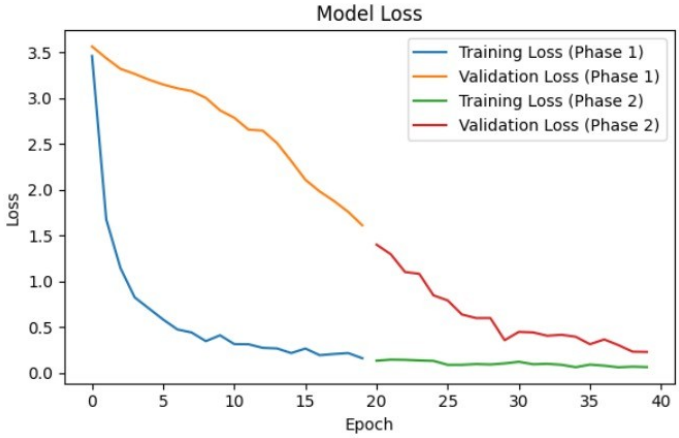


Fig. 4. Model Loss

The fig. 4. illustrating the loss values over 40 epochs for different training phases of a deep learning model. The x-axis represents the number of epochs, while the y-axis shows the loss values. The blue line indicates the training loss for Phase 1, which drops sharply and stabilizes at a low value. The orange line represents the validation loss for Phase 1, showing a gradual decline. In Phase 2, the red line represents the training loss, which starts higher but decreases steadily, while the green line represents the validation loss, which remains consistently low. This graph highlights the model's improvement in minimizing loss over time, indicating effective learning.

V. CONCLUSION

This study presents a deep learning-based approach for handwritten signature verification using transfer learning with the VGG16 model. Signature authentication is essential in security-sensitive areas such as banking and legal documentation, where preventing forgery is crucial. Traditional methods often struggle with variations in signatures and skilled forgeries, making deep learning a viable solution. The proposed model leverages the VGG16 architecture to extract key distinguishing features from handwritten signatures. A well-balanced dataset of genuine and forged signatures is used for training, with pre-processing techniques like grayscale conversion, image resizing, and normalization applied to enhance feature extraction. Data augmentation methods such as rotation and scaling further improve generalization. The verification process involves comparing a new signature against stored references using similarity measures and threshold-based classification. Experimental results show that the model achieves high accuracy in distinguishing genuine signatures from forgeries. Two experiments were conducted: one using a CNN alone and another integrating a pre-trained VGG16 model. The latter outperformed the former, confirming the benefits of transfer learning. The findings highlight the model's reliability and efficiency, making it suitable for real-world applications like financial transactions and identity verification. This approach strengthens security by reducing fraud risks. Future work could explore additional deep learning models and larger datasets to enhance accuracy and robustness further.

REFERENCES

[1] O. D. Akinrolabu, O. O. Ajayi, A. E. Ebitigha, A. A. Adegbite, J. R. Obafemi and J. K. Ogunleye, "Handwritten Signature Verification Model Using Transfer Deep Learning Technique," 2024 International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG), Omu-Aran, Nigeria, 2024, pp. 1-6, doi: 10.1109/SEB4SDG60871.2024.10629731.

[2] P. Blessy, K. Kathiresan and N. Yuvaraj, "Deep Learning Approach to Offline Signature Forgery Prevention," 2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2023, pp. 1570-1575, doi: 10.1109/ICACCS57279.2023.10112906.

[3] B. Mustafa, R. Taha, O. M. Fahmy and S. M. Afifi, "Handwritten Signature Recognition using Deep Learning," 2023 International Conference on Microelectronics (ICM), Abu Dhabi, United Arab Emirates, 2023, pp. 1-6, doi: 10.1109/ICM60448.2023.10378888.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:09 UTC from IEEE Xplore. Restrictions apply.

[4] P. William, G. R. Lanke, S. Pundir, I. Kumar, M. Gupta and S. Shaw, "Implementation of Hand Written based Signature Verification Technology using Deep Learning Approach," 2023 4th International Conference on Intelligent Engineering and Management (ICIEM), London, United Kingdom, 2023, pp. 1-6, doi: 10.1109/ICIEM59379.2023.10167195.
[5] K. Harika, P. Dhanalakshmi, S. S. K. Reddy, S. Shakeer, M. V. S. Thanmayi and V. Haripriya, "Handwritten Signature Recognition using MobileCNN," 2024 International Conference on Intelligent Systems for Cybersecurity (ISCS), Gurugram, India, 2024, pp. 1-7, doi: 10.1109/ISCS61804.2024.10581340.
[6] S. C. Nossam, R. A. Katakam, G. Pulastya and S. Jayan, "Signature Forgery Detection and Verification using Deep Learning Techniques," 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kamand, India, 2024, pp. 1-6, doi: 10.1109/ICCCNT61001.2024.10726260.
[7] A. Kazi, V. Bharadi and K. Prasad, "Deep Learning Based Dynamic Signature Analysis for Multiclass Classification," 2023 6th International Conference on Advances in Science and Technology (ICAST), Mumbai, India, 2023, pp. 256-261, doi: 10.1109/ICAST59062.2023.10454978.
[8] Eman Alajrami, Belal A. M. Ashqar, Bassem S. Abu-Nasser, Ahmed J. Khalil, Musleh M. Musleh, Alaa M. Barhoom, Samy S. Abu-Naser,"Handwritten Signature Verification using Deep Learning", International Journal of Academic Multidisciplinary Research (IJAMR), ISSN: 2643-9670 Vol. 3 Issue 12, December – 2019.
[9] O. AbuAlghanam, L. Albdour, and A. Omar, "Multimodal biometric fusion online handwritten signature verification using neural network and support vector machine," International Journal of Innovative Computing, Information, and Control (ICIC International), vol. 17, no. 5, pp. 1691 1703, 2021.
[10] Djeziri , Salah Guessoum, Ahmed Oussedik, Seifedine. (2013). Off-Line Signature Verification Based on Graph Matching. 978-3-642-41651-6. 10.1007/978-3-642-41652-39.
[11] Malik, A. S., Khokhar, R. H. (2017). Online signature verification using dynamic time warping and neural networks. International Journal of Biometrics, 9(1), 47-63.
[12] Yousefi, S., Iranmehr, A., Nasiri, A. (2013). Off-line signature verification based on HOG features and SVM classifier. Expert Systems with Applications, 40(3), 909-915.
[13] Hafemann, D. R., Oliveira, L. S., Sabourin, R. (2017). Offline signature verification using deep convolutional neural networks. International Joint Conference on Neural Networks (IJCNN), 3279-3286.
[14] Balogun, A. O., Adebayo, O. S., Ojo, J. A. (2021). Handwritten signature verification using transfer learning with deep convolutional neural networks. IEEE Access, 9, 89894-89904.
[15] Chollet, F. (2005). Siamese neural networks for one-shot image recognition. NIPS Workshop on Deep Learning, 1-9.
[16] C. Sharma, "Transfer learning and its application in computer vision: A review," in Conference Paper*, Electrical and Computer Engineering, University of Waterloo, 2022.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 06:59:09 UTC from IEEE Xplore. Restrictions apply.